SimplePH Channel Flow Verification

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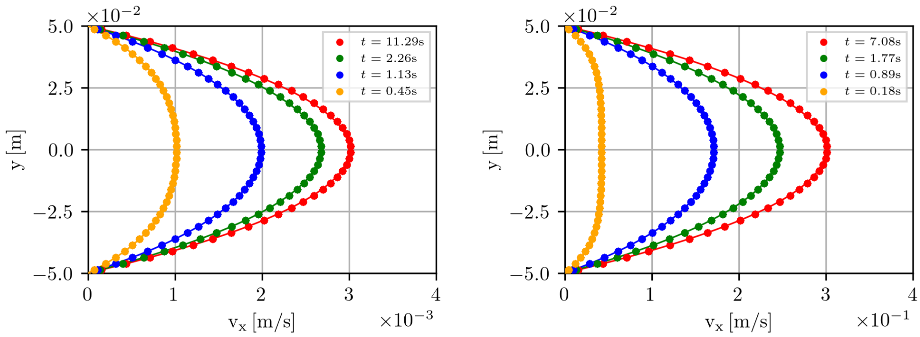
1 Poiseuille Flow

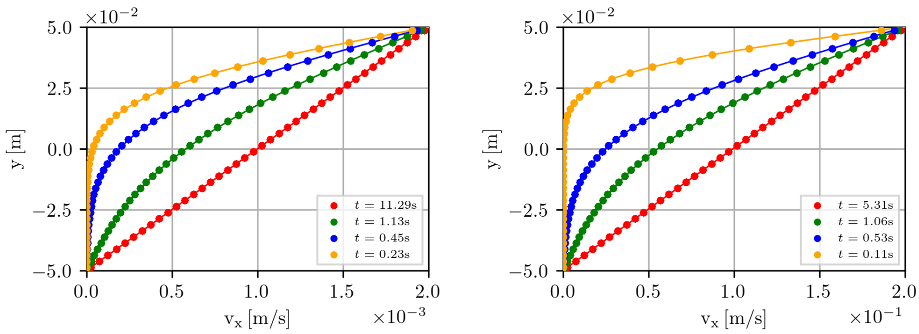


(a) Re = 0.1

(b) Re = 10

Figure 1: Poiseuille velocity profiles compared to analytical solution [2].

2 Couette Flow

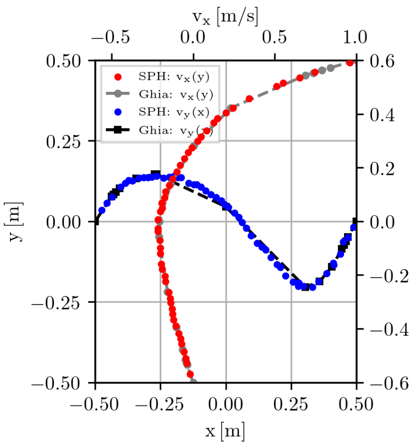


(a) Re = 0.1

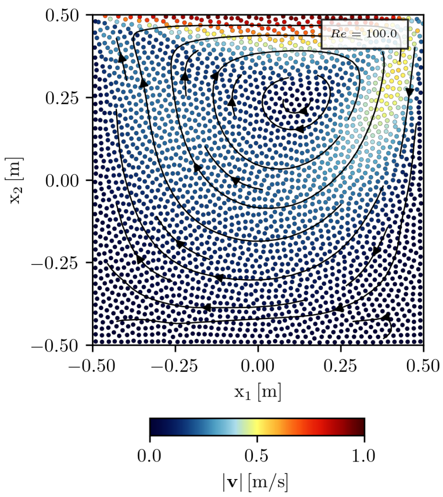
(b) Re = 10

Figure 2: Couette velocity profiles compared to analytical solution [2].

3 Lid Driven Cavity (LDC) Flow

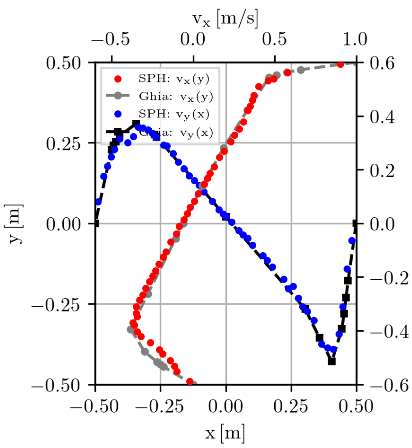


(a) Velocity profiles.

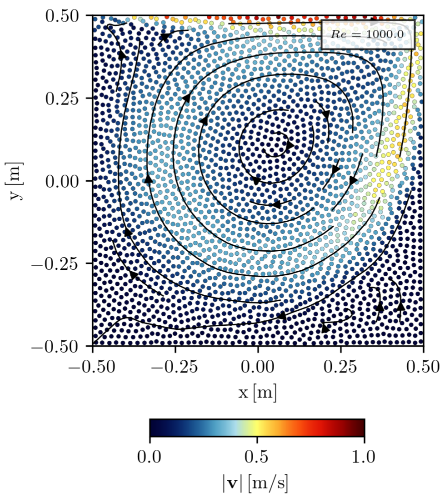


(b) Velocity field.

Figure 3: Lid driven cavity flow velocity field and velocity profiles compared to solution of Ghia et al. [1] for Re = 100.0.



(a) Velocity profiles.



(b) Velocity field.

Figure 4: Lid driven cavity flow velocity field and velocity profiles compared to solution of Ghia et al. [1] for Re = 1000.0.

References

[1] U Ghia, K Ghia, Kirti N, and CT Shin. High-re solutions for incompressible flow using the navier-stokes equations and a multigrid method. *Journal of*

Computational Physics, 48(3):387–411, 1982.

[2] Joseph P Morris, Patrick J Fox, and Yi Zhu. Modeling low Reynolds number incompressible flows using SPH. *Journal of Computational Physics*, 136(1): 214–226, 1997. doi: 10.1006/jcph.1997.5776.